Relational Algebra (RDMS)

SQL —— RA —— Computer

(logical plan) (Physical Plan)

Notation:

1. Project / Column (π)
2. Tuple / Row (σ)
3. Rename (ρ)
4. Assign ($\leftarrow$)

FROM
↓
WHERE
↓
GROUP BY
↓
HAVING
↓
SELECT
↓
ORDER BY

- SELECT id FROM student WHERE marks > 80

$\pi_{id}\left(\sigma_{marks>80}(student)\right)$

$R \leftarrow \sigma_{marks>80}(student)$

$\pi_{id}(R)$

student

id	marks
A	90
B	81
C	79

- SELECT * FROM student WHERE Dept = 'CSE' AND att > 10

$$\pi_{id,\,name,\,Dept,\,att}\left(\sigma_{Dept='CSE'\ AND\ att>10}(student)\right)$$

OR

$$\sigma_{Dept='CSE'\ AND\ att>10}(student)$$

id	name	Dept	att
		CSE	16
		CS	
		CSE	12

student

- Rename id to sid

$\rho_{(sid,name)}(student)$

id	Name

student

- Change table name to s

$\rho_{s(sid,name)}(student)$

- Find $(a+b/2)/(b*0.5)$

$\underbrace{(a+b/2)}_{n} / \underbrace{(b*0.5)}_{d}$

n = a+b/2;

d = b * 0.5;

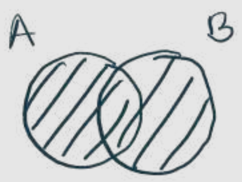
result = n/d;

- Find id of students who got more than 80

$R \leftarrow \sigma_{m>80}(student)$

$\pi_{id}(R)$

Union

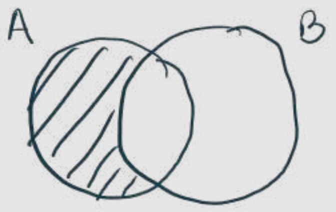


$A \cup B$

A or B

$A \cap B$

A and B



$A - B$

↳ Find people who are not in B.

student

id	name
1	A
2	B
3	C

Grade

id	Ccode	Grade
1	a	A
2	b	A+
3	c	A

Cartesian Product

↳ একটার বেশি Table

$\text{Student} \bowtie_{\text{student.id} = \text{Grade.id}} \text{Grade}$

RA Tree

$\pi_{id}(\sigma_{grade>80}(\text{student}))$

↑ Leaf

π_{id}

|

$\sigma_{grade>80}$

↓

student

Write RA to convert the following SQL queries.

```
15. SELECT distinct SName, dept FROM Student WHERE dept = 'EEE';
16. SELECT * FROM StudentCourse WHERE Grade = 'A';
17. SELECT distinct FIN, FName FROM Faculty, FacultyCourses, Course
    WHERE Faculty.FIN = FacultyCourses.FIN
    AND FacultyCourses.CCode = Course.CCode
    AND Course.CName = 'Automata and Computability';
18. SELECT distinct SID, SName, CName FROM Student, ST, Course
    WHERE Student.SID = ST.SID
    AND ST.assignedCourse = Course.CCode;
```

15 $\pi_{SName, dept}(\sigma_{dept='EEE'}(\text{student}))$

17 $\pi_{FIN, Fname}(\sigma_{CName='AC'}((\text{Faculty} \bowtie_{FIN=FIN} \text{FacultyCourse}) \bowtie_{Ccod=Ccode} \text{Course}))$

18 $\pi_{SID, SName, Cname}((\text{Student} \bowtie_{SID=SID} \text{ST}) \bowtie_{AC=Ccode} \text{Course})$

Example Problems - I

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)
ST(SID, assignedCourse)
Faculty(FIN, FName, LName, Desk#)
Course(CCode, CName)
FacultyCourses(FIN, CCode, Section)
-- FIN references Faculty(FIN), CCode references Course(CCode)

A) Write RA for the following:

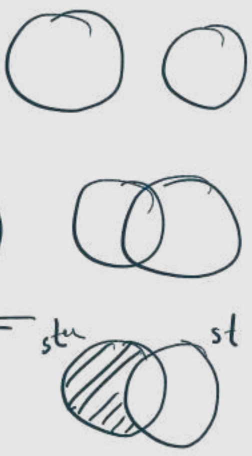
1. Find students with at least 70 credits completed from 'CS' dept.
2. Find FName and Desk# of all faculties.
3. Find SID, SName of Students from 'CSE' dept.
4. Find the names of students who are not an ST.

(1) $\sigma_{C>=70 \text{ AND } Dept='CS'}(student)$

(2) $\pi_{Fname, Desk\#}(Faculty)$

(3) $\pi_{SID, Sname}(\sigma_{D='Cse'}(student))$

(4) $\pi_{name}(student) - \pi_{name}(student \bowtie_{SID=SID} ST)$



Example Problems - II

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)
ST(SID, assignedCourse) -- *SID references Student(SID)*
Faculty(FIN, FName, Desk#)
Course(CCode, CName)
FacultyCourses(FIN, CCode, Section)
-- *FIN references Faculty(FIN), CCode references Course(CCode)*

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle
2. Find ST's SName, dept, assignedCourse.
3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.
4. Find Faculties FName, Desk# for those who are taking either 'Database Systems or 'AI' Courses.

(1) $\rho_{(Code, CTitle)}(Course)$

(2) $\pi_{SName, dept, AC}(ST \bowtie_{Sid=Sid} student)$

(3)

$$R \leftarrow \left(\sigma_{AC = \text{'CSE370'} \text{ OR } AC = \text{'CSE230'}} \left(\text{Faculty} \bowtie_{in=in} \text{Faculty Courses} \right) \right)$$

$$\pi_{\text{Fname}, \text{Desk\#}} (R)$$

(d)

$$\pi_{\text{Fname}, \text{Desk\#}} \left((\text{Faculty} \bowtie_{FIN=FIN} \text{FacultyCourses}) \bowtie_{Ccode=Ccode} \text{Course} \right)$$

RA Tree

① $\pi_{SName}(\text{Student}) - \pi_{SName}(\text{Student} \bowtie_{SID=SID} \text{ST})$

Ans:

```
                 −
               /   \
         πSName    πSName
           |          |
        ⋈SID=SID    Student
        /      \
      ST      Student
```

② $\pi_{SID,\ SName,\ CName}((\text{Student} \bowtie_{SID=SID} \text{ST}) \bowtie_{AssignedCourse=CCode} \text{Course})$

Ans:

```
              πSID, SName, CName
                      |
                  ⋈SID=SID
                  /        \
  ⋈AssignedCourse=CCode    Student
     /        \
  Course       ST
```